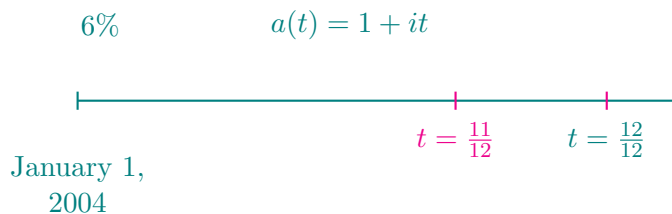


Problem 4.15

A deposit is made on January 1, 2004. The investment earns 6% simple interest. Calculate the monthly effective interest rate for the month of December 2004.

Solution



For simple interest,

$$a(t) = 1 + it,$$

with

$$i = 0.06.$$

The monthly effective interest rate for December 2004 is

$$i_{12} = \frac{a(12) - a(11)}{a(11)}.$$

Since time is measured in years, December begins at

$$t = \frac{11}{12}$$

and ends at

$$t = \frac{12}{12} = 1.$$

So,

$$a\left(\frac{11}{12}\right) = 1 + 0.06\left(\frac{11}{12}\right) = 1.055,$$

and

$$a\left(\frac{12}{12}\right) = 1 + 0.06\left(\frac{12}{12}\right) = 1.06.$$

Therefore,

$$i_{12} = \frac{1.06 - 1.055}{1.055} = 0.0047393.$$

Hence, the monthly effective interest rate for December 2004 is

$$\boxed{0.47393\%}.$$